

The Maximum Principle for the Heat Equation

Math Department

The Heat Equation

We consider the heat equation on a spatial interval:

$$u_t = k u_{xx}$$

where

- ▶ $u(x, t)$ = temperature
- ▶ x = spatial variable
- ▶ t = time
- ▶ $k > 0$ = thermal diffusivity

Domain:

$$a \leq x \leq b, \quad 0 \leq t \leq T$$

Initial condition:

$$u(x, 0) = f(x)$$

Boundary conditions:

$$u(a, t) = g_1(t), \quad u(b, t) = g_2(t)$$

The Parabolic Boundary

The domain in the (x, t) plane is a rectangle.

Interior:

$$a < x < b, \quad 0 < t \leq T$$

Parabolic boundary consists of

$$t = 0, \quad x = a, \quad x = b$$

but **not the top edge** $t = T$.

This reflects the direction of time evolution in the heat equation.

The Maximum Principle

Maximum Principle

Suppose $u(x, t)$ satisfies

$$u_t = ku_{xx}$$

in the interior of the rectangle.

Then the maximum value of u on the closed region

$$a \leq x \leq b, \quad 0 \leq t \leq T$$

occurs on the **parabolic boundary**

$$t = 0, \quad x = a, \quad x = b.$$

In particular, an interior maximum cannot occur unless u is constant.

Idea of the Proof

Suppose u attains a maximum at an interior point

$$(x_0, t_0)$$

with

$$a < x_0 < b, \quad 0 < t_0 \leq T$$

At a local maximum we must have

$$u_x(x_0, t_0) = 0$$

$$u_{xx}(x_0, t_0) \leq 0$$

and

$$u_t(x_0, t_0) \geq 0$$

Using the PDE

From the heat equation

$$u_t = k u_{xx}$$

evaluate at the maximum point:

$$u_t(x_0, t_0) = k u_{xx}(x_0, t_0)$$

But

$$u_{xx}(x_0, t_0) \leq 0$$

so

$$u_t(x_0, t_0) \leq 0$$

Yet the maximum assumption gave

$$u_t(x_0, t_0) \geq 0$$

Conclusion of the Argument

The two inequalities imply

$$u_t(x_0, t_0) = 0$$

and

$$u_{xx}(x_0, t_0) = 0$$

This forces the solution to be locally flat.

By repeating the argument one shows the solution must be constant in the connected region.

Therefore a nonconstant solution cannot achieve a maximum in the interior.

Maximum Principle

Thus:

$$\max_{a \leq x \leq b, 0 \leq t \leq T} u(x, t)$$

occurs on the boundary

$$t = 0, \quad x = a, \quad x = b.$$

Similarly,

$$\min_{a \leq x \leq b, 0 \leq t \leq T} u(x, t)$$

also occurs on the boundary.

Temperature extremes are controlled by initial and boundary data.

Physical Interpretation

The maximum principle reflects the physics of heat flow.

Heat diffuses from hotter regions to colder ones.

Therefore

- ▶ temperature peaks smooth out
- ▶ new interior maxima cannot appear

The hottest point must originate from the initial or boundary data.

Important Consequence: Uniqueness

Suppose u and v are two solutions with the same data.
Define

$$w = u - v$$

Then

$$w_t = kw_{xx}$$

with zero initial and boundary conditions.

By the maximum principle,

$$w \equiv 0$$

Thus

$$u = v$$

Summary

For the heat equation:

$$u_t = ku_{xx}$$

- ▶ Interior maxima cannot occur.
- ▶ Extreme values occur on the parabolic boundary.
- ▶ The principle implies uniqueness of solutions.
- ▶ Physically it expresses the smoothing nature of heat diffusion.